IDS Prediction Dashboard Report

Internship Learning Documentation & Technical Summary

1. Executive Summary

This project outlines the implementation of a local Intrusion Detection System (IDS) Prediction Dashboard designed to classify network traffic flows and identify potential cyberattack patterns. The system leverages a machine learning classifier to analyze flow traffic data uploaded in CSV format. Predictions are labeled by threat type and categorized into severity levels (Normal vs Alert) for administrative review. The primary goal of this tool is to provide security personnel with a lightweight, reusable, and interactive interface to analyze logs offline and visual distribution counts of network traffic threats.

2. Classifier Engine Specifications

The machine learning subsystem uses a **Random Forest Classifier** trained on synthetic network flow logs designed to mimic standard threat patterns (e.g. CICIDS2017 datasets). The model evaluates logs against 10 core traffic attributes. Prediction probabilities are output for 7 classes:

- **BENIGN**: Standard, harmless system operations.
- **DoS (Denial of Service)**: Volumetric traffic spikes flooding network links.
- **Brute Force**: Repetitive credential login attempts flagged on Ports 21/22.
- **SQL Injection**: Malicious structured queries targeting data entry points.
- **XSS (Cross-Site Scripting)**: Payload patterns injected via web requests.
- **PortScan**: Multi-port probes scanning for open target sockets.
- **Infiltration**: Vulnerability exploits attempting unauthorized host takeovers.

3. Severity Mapping Policy

The dashboard translates categorical predictions into risk profiles to highlight items requiring immediate triage:

Predicted Label	Mapped Severity	Administrative Action
BENIGN	Normal	None. Logged for standard record maintenance.
DoS / Brute Force	Alert	Immediate threat mitigation. Check traffic rate limit and firewall rules.
SQL Injection / XSS	Alert	App vulnerability triage. Inspect web app input sanitizers.
PortScan / Infiltration	Alert	System audit. Investigate host integrity and restrict open ports.

4. Input Feature Requirements

The IDS Dashboard validates uploaded files to ensure they conform to the features used in model training. The table below defines the 10 selected network metrics required in the CSV uploader:

Feature Name	Data Type	Description / Significance
Destination Port	Integer	Target socket identifier (determines application channel).
Flow Duration	Float	Total duration of the network connection stream (microseconds).
Total Fwd Packets	Integer	Total count of packets sent in the forward direction.
Total Backward Packets	Integer	Total count of packets sent in the backward direction.
Fwd Packet Length Max	Float	Maximum length of forward packets in bytes.
Bwd Packet Length Max	Float	Maximum length of backward packets in bytes.
Flow Bytes/s	Float	Rate of byte transmission per second.
Flow Packets/s	Float	Rate of packet transmissions per second.
Packet Length Mean	Float	Average size of packets transmitted in the stream.
Average Packet Size	Float	Mean packet size adjusted for directional headers.

5. Dashboard Capabilities & Usability

The interface was designed in Streamlit for optimal local deployment, responsiveness, and clear presentation. It provides safety layers and clear user prompts:

- **Validation Engine**: Ensures uploaded data contains all 10 feature headers. Displays descriptive errors detailing exactly what is missing without crashing.
- **Summary Indicators**: Displays critical aggregate metrics (Total Logs, Normal Count, Attack Count, Top Attack Type) in styled glassmorphic widgets.
- **Interactive Data Plotting**: Utilizes Plotly components to display class ratios, normal-to-alert proportions, and multidimensional cluster scatter plots.
- **Structured Downloads**: Outputs a clean file mapping row IDs to predictions, severities, and timestamps conforming strictly to project submission requirements.

6. Project Setup Summary

To run the dashboard locally, unzip the project folder and run the following in your terminal:
pip install -r requirements.txt
python generate_assets.py (Optional: regenerates the ML models and data)
streamlit run app.py